



Interval Estimates

In many practical situations, the population parameter is unknown.

Point and Interval Estimates

Point estimates are statistics, such as $\hat{p}$ and $\bar{x}$, that are used to estimate population parameters.

- The point estimate for p is $\hat{p}$.
- The point estimate for μ is $\bar{x}$.

► **Remember** p represents a population proportion, and $\hat{p}$ represents a sample proportion. μ represents a population mean, and $\bar{x}$ represents a sample mean.

An **interval estimate** is a range of values that contains the point estimate and, therefore, is likely to contain the population parameter. It is determined by the **margin of error**, the greatest likely difference between the point estimate and the parameter.

Find the lower limit by subtracting the margin of error from the point estimate. Find the upper limit by adding the margin of error to the point estimate. Look at the three ways you may see interval estimates described.

General form	Example
point estimate $\pm$ margin of error	$14\% \pm 2\%$
between lower limit and upper limit	between 12% and 16%
(lower limit, upper limit)	(12%, 16%)

Using Interval Estimates: Proportions

A survey determined that the proportion, p , of voters in a city who support a local wildlife refuge is likely to be in the interval $(0.62, 0.67)$. Write the interval estimate in the form *point estimate* $\pm$ *margin of error*.

The margin of error is half the difference of the upper and lower limits.

$$\frac{0.67 - 0.62}{2} = 0.025$$

To find the point estimate, $\hat{p}$, add the margin of error to the lower limit.

$$0.62 + 0.025 = 0.645$$

The interval estimate is 0.645 ± 0.025 .

If 750 voters are randomly sampled, what is the best estimate for the number of voters who support the wildlife refuge?

Because $\hat{p}$ is the best estimate of p , then 64.5% of the 750 voters, or about 484 of the voters, are likely to support the refuge.

► Q&A

Q What is the interval for the number of voters who are likely to support the refuge?

A (465, 503)

Using Interval Estimates: Means

A study of a sample of two-toed sloths in a given area indicates that the mean weight of all the sloths in that area is between 16.8 lb and 17.6 lb. Write the interval estimate in the form *point estimate* $\pm$ *margin of error*.

Because $\frac{17.6 - 16.8}{2} = 0.4$, the margin of error is ± 0.4 lb and the point estimate $\bar{x}$ is 16.8 lb + 0.4 lb, or 17.2 lb. The interval estimate is 17.2 lb $\pm$ 0.4 lb.

- **Think About It** Sloths can be two-toed or three-toed and are most commonly known for their slowness.

Using Point Estimates: Proportions

In a random sample of 63 homeowners in a city, 15 homeowners said that they would support a ban on all nonnatural lawn fertilizers in an effort to protect the fish in the local waterways. The sampling method had a margin of error of $\pm 3.5\%$. Write the interval estimate for the proportion that would support the ban in the form (*lower limit*, *upper limit*).

First determine the point estimate.

$$\hat{p} = \frac{15}{63} \approx 0.238 = 23.8\%$$

Then subtract the margin of error from the point estimate to get the lower limit and add the margin of error to the point estimate to get the upper limit.

$$\text{lower limit: } 23.8\% - 3.5\% = 20.3\%$$

$$\text{upper limit: } 23.8\% + 3.5\% = 27.3\%$$

The interval estimate is (20.3%, 27.3%).

In a random sample of 120 homeowners in the city, how many homeowners would you expect to say they would support the ban? Give the answer as an interval.

Because 20.3% of 120 is about 24, and 27.3% is about 33, you would expect between 24 and 33 of the homeowners to say they would support the ban.

- **Think About It** The margin of error is affected by the size of the sample chosen. A surveyor can reduce the margin of error by increasing the number of members in the sample.

Using Point Estimates: Means

In a two-toed sloth study, the sloths in the sample had a mean body length of 63 cm. The study had a margin of error of ± 0.7 cm. Write the interval estimate for the mean body length in the form (*lower limit*, *upper limit*).

$$\text{lower limit: } 63 \text{ cm} - 0.7 \text{ cm} = 62.3 \text{ cm}$$

$$\text{upper limit: } 63 \text{ cm} + 0.7 \text{ cm} = 63.7 \text{ cm}$$

The interval estimate is (62.3 cm, 63.7 cm).

► **Think About It** The interval (62.3 cm, 63.7 cm) means that it is likely, but not certain, that the mean length of all the sloths in the area is between 62.3 cm and 63.7 cm.